

PLANETVORTEX

PLANETVORTEX — Theorem B — the seven equal cells

The theorem edition of the planetary gravity-geometry bench — mini-repository planetvortex/ of the TRIVORTEX framework, version 1.0.0. This dossier binds one statement to its registered protocol; the full science lives in the big monograph (docs/monograph/).

Statement

Place seven equal vortices Γ at the heptagon vertices. Then every Fano line-cell — a three-vortex Trivortex cell in the scalene regime — carries exactly the same Hamiltonian $H_{cell} = -\frac{\Gamma^2}{2\pi} \ln(\sqrt{7} R^3) = -0.154851\dots$, the same angular impulse $I_{cell} = 3\Gamma R^2$ and the same invariant impulse magnitude $|P+iQ|$; the seven cells are congruent as dynamical systems, and their shape cycles are periodic with one common period $T_{shape} = 9.3814\dots$, measured with spread exactly 0.0.

Proof

Each line is a rotation image of L_0 , so the cells are congruent and the equality of invariants follows. The Hamiltonian of one cell is $-\frac{\Gamma^2}{2\pi}(\ln s_1 + \ln s_2 + \ln s_3) = -\frac{\Gamma^2}{2\pi} \ln(R^3 \sqrt{7})$ by the product identity of Theorem A — the same for all seven. The moment of inertia is $\Gamma \sum |z_k|^2 = 3\Gamma R^2$: every vertex sits on the circumcircle. The shape periodicity is the classical integrability of the three-vortex problem: the shape variable moves on the closed level curve of H inside the fixed- I shape domain. The scalene cells are NOT relative equilibria — the maximal shape departure is 0.288 — the symmetry $\text{PSL}(2,7)$ buys is the equality of the cycles, not rigidity. ▢

Protocol binding

item	value
stage	P6
recorded registers	9.1×10^{-14} / spread 0.0
protocol file	results/protocols/ (default preset)
test guard	tests/ re-derives every preset-independent register

Cross-references

- the parent framework: Theorem 3.1 and the three-vortex anchor;
- the sibling bench polyvortex: the W2/W4 registers at $N=7$ — the same heptagon, cross-pinned bit-for-bit;
- the big monograph: Sections 3 (Theorem A), 6 (Theorem C), 7 (Theorem B), 4 (Lemma D), 8 (Lemma E);
- the formalization queue: roadmap item V4 (Coq / Lean 4).